

Module 5: Transformers & Generation Evaluation

Self-Attention, Multi-Head Attention, BERT vs GPT Architectures & BLEU / ROUGE Math

☰ **Module V: NLP Applications, Dialogue Systems & AI Ethics — Complete 10-Topic Tracker**

- Topic 44:** Text Classification (Document Categorization & Naive Bayes)

Topic 46: Named Entity Recognition (NER Sequence Labeling vs. POS) [UPLOADED PYQ]

Topic 48: Rule-Based Machine Translation (Direct, Transfer & Interlingua)

Topic 50: Neural Machine Translation (End-to-End Transformer Seq2Seq)

Topic 52: Bias, Fairness & Stereotyping in Large Language Models
- Topic 45:** Sentiment Analysis (Document, Sentence & Aspect-Based)

Topic 47: Machine Translation Overview & The Vauquois Triangle

Topic 49: Statistical Machine Translation (SMT Noisy Channel Formulation)

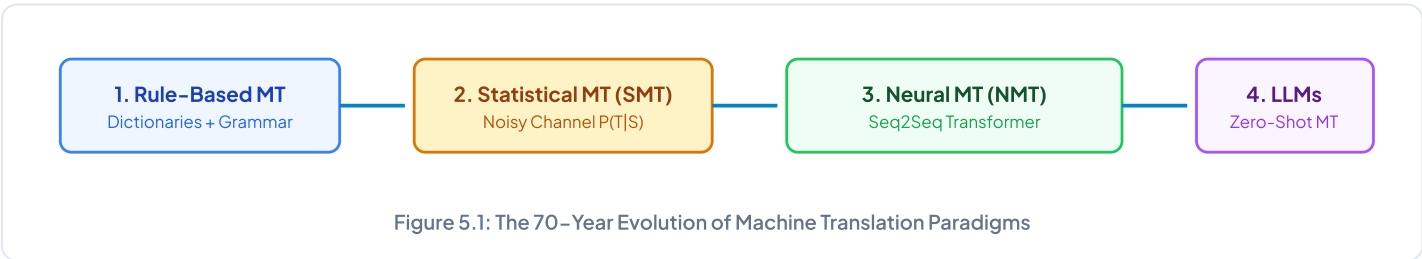
Topic 51: Chatbots & Conversational Dialogue Systems (NLU-DM-NLG Pipeline)

Topic 53: Model Explainability & Interpretability (Attention Heatmaps & LIME)

Topic 44 – 46: Text Classification, Sentiment Analysis & NER [UPLOADED PYQ]

Feature	Part-of-Speech (POS) Tagging	Named Entity Recognition (NER) [UPLOADED PYQ — CS633]
Objective	Assigns syntactic grammatical word classes ('NN', 'VB', 'JJ', 'DT').	Detects and classifies real-world semantic entity mentions.
Granularity	Every single token in the sentence receives a tag.	Only specific entity spans receive labels ('PERSON', 'ORG', 'LOC', 'DATE'). Non-entities are tagged 'O'.
Example	"Sundar [NNP] Pichai [NNP] leads [VBZ] Google [NNP]"	"Sundar Pichai [B-PER, I-PER] leads Google [B-ORG]" (BIO tagging scheme).

Topic 47 – 50: Machine Translation (MT) Paradigm Evolution



Statistical Machine Translation (Noisy Channel Model):

$$\hat{T} = \arg \max_T P(T | S) = \arg \max_T \left[\underbrace{P(S | T)}_{\text{Translation Model (Fidelity)}} \times \underbrace{P(T)}_{\text{Language Model (Fluency)}} \right]$$

Topic 51: Chatbots & Conversational Dialogue Architectures

The 3-Tier Task-Oriented Dialogue System Architecture

1. **Natural Language Understanding (NLU):** Converts user utterance into structured semantic frame: Intent Classification (e.g., `BookFlight`) + Slot Filling (e.g., `Origin: "Ranchi"`, `Destination: "Delhi"`).
2. **Dialogue State Tracker (DST) & Policy Manager:** Maintains conversation history state and selects the next system action (e.g., `AskDepartureDate` or `ExecuteBookingAPI`).
3. **Natural Language Generation (NLG):** Synthesizes natural, fluent response text from the structured action template.

Topic 52 & 53: Ethics, Bias Mitigation & Model Explainability

Ethical / Interpretability Challenge	Manifestation in Language Models	Engineering Mitigation Method
1. Dataset & Representation Bias	Stereotypical associations in word embeddings (e.g., "doctor" – "man" + "woman" $\approx$ "nurse").	Counterfactual data augmentation, debiasing projection subspaces.
2. Model Hallucination	Generating plausibly sounding but factually incorrect assertions with high confidence.	Retrieval-Augmented Generation (RAG), RLHF with truthfulness reward models.
3. Black-Box Interpretability	Complex billions-parameter Transformer decisions cannot be audited for high-stakes healthcare/law.	LIME (Local Interpretable Model-agnostic Explanations), Integrated Gradients, Cross-Attention Heatmaps.

M5 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Compare Rule-Based, Statistical, and Neural Machine Translation systems across 4 dimensions. (8 Marks)

1. **Knowledge Representation:** RBMT relies on explicit handwritten grammar rules and bilingual dictionaries; SMT uses phrase-based translation tables; NMT uses continuous distributed vector representations in a unified neural network.
2. **Fluency & Context:** RBMT produces robotic, literal translations; SMT handles local phrases well but struggles with long-distance reordering; NMT generates highly fluent translations with global sentence context.
3. **Development Cost:** RBMT requires decades of expert linguistic labor; SMT/NMT learn automatically from parallel sentence corpora.
4. **Hardware Requirements:** RBMT runs with low compute; NMT requires massive GPU accelerators for Transformer matrix multiplications.